

Goal

- Develop a structure that will allow user to insert/delete/find records in
constant average time
 - structure will be a table (relatively small)
 - table completely contained in memory
 - implemented by an array
 - capitalizes on ability to access any element of the array in *constant* time

Hash Function

- Determines position of key in the array.
- Assume table (array) size is N
- Function $f(x)$ maps any key x to an int between 0 and $N-1$
- For example, assume that $N=15$, that key x is a non-negative integer between 0 and MAX_INT , and hash function $f(x) = x \% 15$.
- (Hash functions for strings aggregate the character values --- see Weiss §5.2.)

Hash Function

Let $f(x) = x \% 15$. Then,

if $x =$	25	129	35	2501	47	36
$f(x) =$	10	9	5	11	2	6

Storing the keys in the array is straightforward:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
—	—	47	—	—	35	36	—	—	129	25	2501	—	—	—

Thus, *delete* and *find* can be done in $O(1)$, and
also *insert*, except...

Hash Function

What happens when you try to insert: $x = 65$?

$$x = 65$$

$$f(x) = 5$$

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
—	—	47	—	—	35	36	—	—	129	25	2501	—	—	—
					65 (?)									

This is called a *collision*.

Handling Collisions

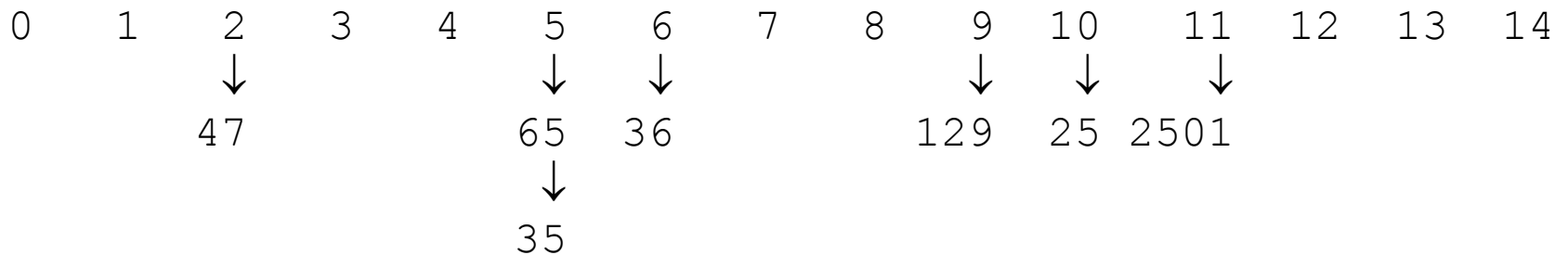
- Separate Chaining
- Open Addressing
 - Linear Probing
 - Quadratic Probing
 - Double Hashing

Handling Collisions

Separate Chaining

Separate Chaining

Let each array element be the head of a chain.

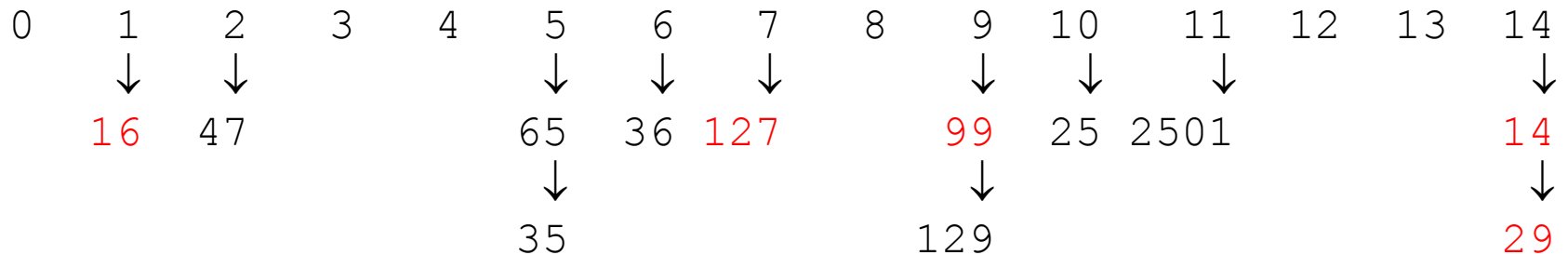


Where would you store: 29, 16, 14, 99, 127 ?

Separate Chaining

Let each array element be the head of a chain:

Where would you store: 29, 16, 14, 99, 127 ?



New keys go at the front of the relevant chain.

Separate Chaining: Disadvantages

- Parts of the array might never be used.
- As chains get longer, search time increases to $O(n)$ in the worst case.
- Constructing new chain nodes is relatively expensive (still constant time, but the constant is high).
- Is there a way to use the “unused” space in the array instead of using chains to make more space?

Handling Collisions

Linear Probing

Linear Probing

Let key x be stored in element $f(x)=t$ of the array

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
		47			35	36			129	25	2501			
					65 (?)									

What do you do in case of a collision?

If the hash table is *not full*, attempt to store key in the next array element (in this case $(t+1)\%N$, $(t+2)\%N$, $(t+3)\%N$...)

until you find an empty slot.

Linear Probing

Where do you store 65 ?

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
		47			35	36	65		129	25	2501			
					↑	↑	↑							
					attempts									

Where would you store: 29?

Linear Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1)\%N$, $(t+2)\%N$, ...

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
		47			35	36	65		129	25	2501			29
														↑
														attempts

Where would you store: 16?

Linear Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1)\%N$, $(t+2)\%N$, ...

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	16	47			35	36	65		129	25	2501			29
	↑													

Where would you store: 14?

Linear Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1)\%N$, $(t+2)\%N$, ...

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
14	16	47			35	36	65		129	25	2501			29
↑														↑
														attempts

Where would you store: 99?

Linear Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1)\%N$, $(t+2)\%N$, ...

[illegible]

Where would you store: 127 ?

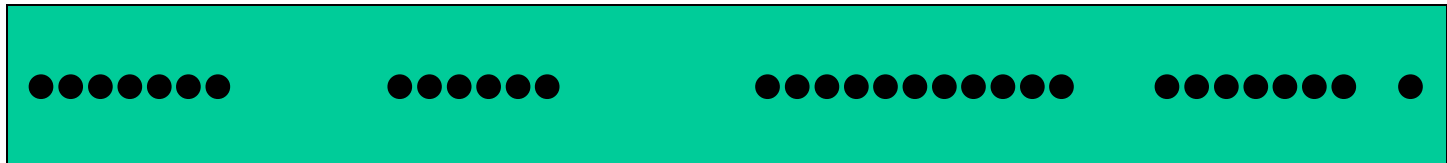
Linear Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1)\%N$, $(t+2)\%N$, ...

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
14	16	47			35	36	65	127	129	25	2501	99		29
							↑	↑						
							attempts							

Linear Probing

- Eliminates need for separate data structures (chains), and the cost of constructing nodes.
- Leads to problem of clustering. Elements tend to *cluster* in dense intervals in the array.



- Search efficiency problem remains.
- Deletion becomes trickier....

Deletion problem

- $H = \text{KEY} \bmod 10$
- Insert 47, 57, 68, 18, 67
- Find 68
- Find 10
- Delete 47
- Find 57

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	

Deletion Problem -- SOLUTION

- “Lazy” deletion
- Each cell is in one of 3 possible states:
 - active
 - empty
 - deleted
- For Find or Delete
 - only stop search when EMPTY state detected (not DELETED)

Deletion-Aware Algorithms

- Insert

- Cell empty or deleted
- Cell active

insert at H, *cell = active*

$H = (H + 1) \bmod TS$

- Find

- cell empty
- cell deleted
- cell active

NOT found

$H = (H + 1) \bmod TS$

if $key == key \text{ in cell} \rightarrow \text{FOUND}$

else $H = (H + 1) \bmod TS$

- Delete

- cell active; $key \neq key \text{ in cell}$
- cell active; $key == key \text{ in cell}$
- cell deleted
- cell empty

$H = (H + 1) \bmod TS$

DELETE; *cell=deleted*

$H = (H + 1) \bmod TS$

NOT found

Handling Collisions

Quadratic Probing

Quadratic Probing

Let key x be stored in element $f(x)=t$ of the array

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
		47			35	36			129	25	2501			
					65 (?)									

What do you do in case of a collision?

If the hash table is *not full*, attempt to store key in array elements $(t+1^2)\%N$, $(t+2^2)\%N$, $(t+3^2)\%N$... until you find an empty slot.

Quadratic Probing

Where do you store **65** ? $f(65)=t=5$

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
		47			35	36			129	25	2501			65
					↑	↑			↑					↑
					t	t+1			t+4					t+9
					attempts									

Where would you store: 29?

Quadratic Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1^2)\%N$, $(t+2^2)\%N$...

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
29		47			35	36			129	25	2501			65
↑														↑
t+1														t
														attempts

Where would you store: 16?

Quadratic Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1^2)\%N$, $(t+2^2)\%N$...

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
29	16	47			35	36			129	25	2501			65
	↑													
	t													
attempts														

Where would you store: 14?

Quadratic Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1^2)\%N$, $(t+2^2)\%N$...

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
29	16	47	14		35	36			129	25	2501			65
↑			↑											↑
t+1			t+4											t
														attempts

Where would you store: 99?

Quadratic Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1^2)\%N$, $(t+2^2)\%N$...

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
29	16	47	14		35	36			129	25	2501		99	65
									↑	↑			↑	
									t	t+1			t+4	
									attempts					

Where would you store: 127 ?

Quadratic Probing

If the hash table is *not full*, attempt to store key in array elements $(t+1^2)\%N$, $(t+2^2)\%N$...

Where would you store: 127 ?

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
29	16	47	14		35	36	127		129	25	2501		99	65
							↑							
							t							
							attempts							

Quadratic Probing

- Tends to distribute keys better than linear probing
- Alleviates problem of clustering
- Runs the risk of an infinite loop on insertion, unless precautions are taken.
- E.g., consider inserting the key 16 into a table of size 16, with positions 0, 1, 4 and 9 already occupied.
- Therefore, table size should be prime.

Handling Collisions

Double Hashing

Double Hashing

- Use a hash function for the decrement value
 - $\text{Hash}(\text{key}, i) = H_1(\text{key}) - (H_2(\text{key}) * i)$
- Now the decrement is a function of the key
 - The slots visited by the hash function will vary even if the initial slot was the same
 - Avoids clustering
- Theoretically interesting, but in practice slower than quadratic probing, because of the need to evaluate a second hash function.

Double Hashing

Let key x be stored in element $f(x)=t$ of the array

Array:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
		47			35	36			129	25	2501			
					65 (?)									

What do you do in case of a collision?

Define a second hash function $f_2(x)=d$. Attempt to store key in array elements $(t+d)\%N$, $(t+2d)\%N$, $(t+3d)\%N$...

until you find an empty slot.

Double Hashing

- Typical second hash function

$$f_2(x) = R - (x \% R)$$

where R is a prime number, $R < N$

Double Hashing

Where do you store **65** ? $f(65)=t=5$

Let $f_2(x) = 11 - (x \% 11)$ $f_2(65)=d=1$

Note: $R=11, N=15$

Attempt to store key in array elements $(t+d)\%N$,
 $(t+2d)\%N, (t+3d)\%N \dots$

Array:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
		47			35	36	65		129	25	2501			
					↑	↑	↑							
					t	t+1	t+2							
					attempts									

Double Hashing

If the hash table is *not full*, attempt to store key in array elements $(t+d)\%N$, $(t+2d)\%N$...

Let $f_2(x) = 11 - (x \% 11)$ $f_2(29) = d = 4$

Where would you store: 29?

Array:

[illegible]

Double Hashing

If the hash table is *not full*, attempt to store key
in array elements $(t+d)\%N$, $(t+2d)\%N$...

Let $f_2(x) = 11 - (x \% 11)$ $f_2(16) = d = 6$

Where would you store: 16?

Array:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	16	47			35	36	65		129	25	2501			29
	↑													
	t													
attempt														

Where would you store: 14?

Double Hashing

If the hash table is *not full*, attempt to store key
in array elements $(t+d)\%N$, $(t+2d)\%N$...

Let $f_2(x) = 11 - (x \% 11)$ $f_2(14) = d = 8$

Array:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
14	16	47			35	36	65		129	25	2501			29
↑							↑							↑
t+16							t+8							t
attempts														

Where would you store: 99?

Double Hashing

If the hash table is *not full*, attempt to store key
in array elements $(t+d)\%N$, $(t+2d)\%N$...

Let $f_2(x) = 11 - (x \% 11)$ $f_2(99) = d = 11$

Array:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
14	16	47			35	36	65		129	25	2501	99		29
	↑				↑				↑			↑		
	t+22				t+11				t			t+33		
attempts														

Where would you store: 127 ?

Double Hashing

If the hash table is *not full*, attempt to store key
in array elements $(t+d)\%N$, $(t+2d)\%N$...

Let $f_2(x) = 11 - (x \% 11)$ $f_2(127) = d = 5$

Array:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
14	16	47			35	36	65		129	25	2501	99		29
		↑					↑					↑		
		t+10					t					t+5		
attempts														

Infinite loop!

REHASHING

- When the load factor exceeds a threshold, double the table size (smallest prime $> 2 * \text{old table size}$).
- Rehash each record in the old table into the new table.
- Expensive: $O(N)$ work done in copying.
- However, if the threshold is large (e.g., $\frac{1}{2}$), then we need to rehash only once per $O(N)$ insertions, so the cost is “amortized” constant-time.

Factors affecting efficiency

- Choice of hash function
 - Collision resolution strategy
 - Load Factor
-
- Hashing offers excellent performance for insertion and retrieval of data.

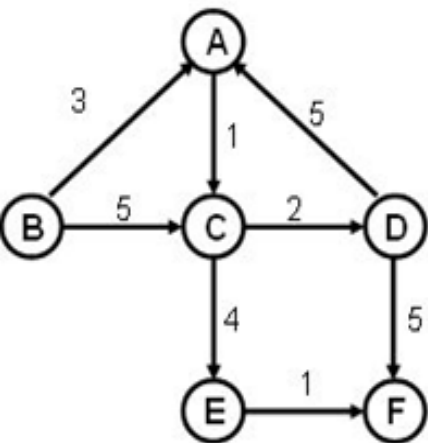
Comparison of Hash Table & BST

	BST	HashTable
Average Speed	$O(\log_2 N)$	$O(1)$
Find Min/Max	Yes	No
Items in a range	Yes	No
Sorted Input	Very Bad	No problems

Use HashTable if there is any suspicion of SORTED input & NO ordering information is required.

Example

Tracing Dijkstra's algorithm starting at vertex B:



Pass:	initially	1	2	3	4	5	6	Shortest distance	Predecessor
Active vertex:		B	A	C	D	E	F		
A	∞	3						3	B
B	0							0	-
C	∞	5	4					4	A
D	∞	∞	∞	6				6	C
E	∞	∞	∞	8	8			8	C
F	∞	∞	∞	∞	11	9		9	E